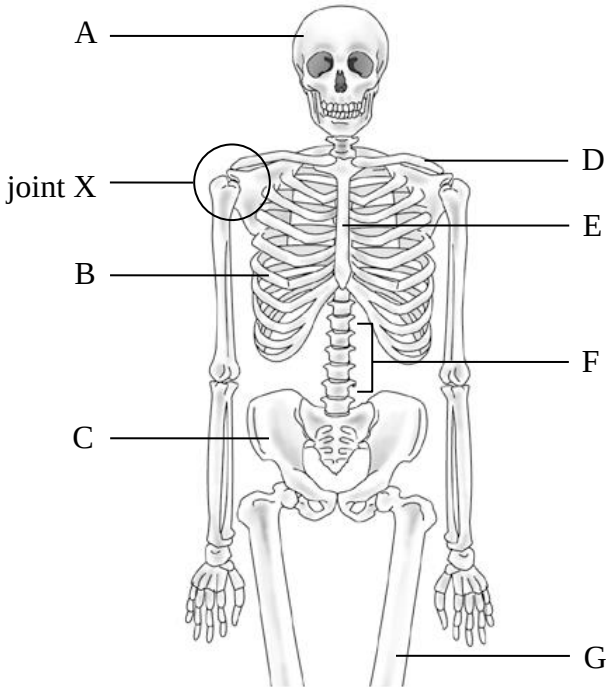
Structured questions

|!|BCELC0180000001|!|

The diagram below shows parts of the human skeleton.



(a) State the ***two*** sub-divisions of the skeleton and classify structures A to F into the two sub-divisions. (4 marks)

(b) What organs do structures B, E and F together protect? (2 marks)

(c) State how the small bones of structure F helps maintain the upright posture of a person. (2 marks)

(d) (i) Name the type of movable joint to which X belong. (1 mark)

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(ii) Name ***two*** components in the joint found at X that can reduce friction in the movement of its bones. (2 marks)

_____

_____

##

(a) Axial skeleton: A, B, E, F (1, 1)

Appendicular skeleton: C, D, G (1, 1)

(b) Heart (1)

Lungs (1)

(c) Any ***two*** of the following: (2)

- They are linked together by ligaments (1)
- to form a column (1)
- and they allow the attachment of muscles for maintaining the posture (1)

(d) (i) Ball-and-socket joint (1)

(ii) Cartilage (1)

Synovial fluid (1)

(11 marks)

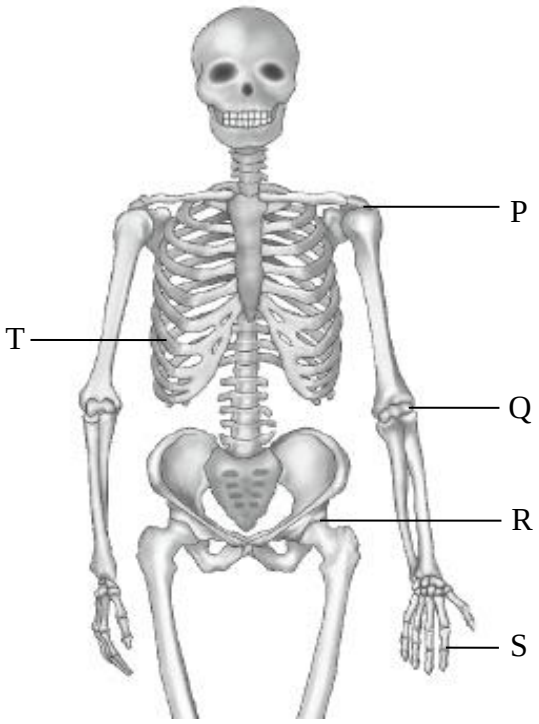
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
	Analysing data	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)(i)	Analysing data	1
(d)(ii)	Understanding basic concepts	2

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|!|BCELC0180000002|!|

The diagram below shows part of the human skeleton.



(a) Joints P, Q, R and S can be classified into two types according to the degree of movement they allow. Name these two types of joints and classify joints P, Q, R and S into these two types. (4 marks)

(b) Give ***three*** functions of structure T. (3 marks)

(c) Osteoporosis is a condition in which the bone density decreases and the bone become weak and brittle. Suggest, with a reason, ***one*** way in which people can reduce the risk of osteoporosis. (2 marks)

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##

(a) Ball-socket-joint: P and R (1, 1)

Hinge joint: Q and S (1, 1)

(b) Any ***three*** of the following: (3)

- protecting the hearts and the lungs /
- assisting ventilation /
- giving shape to the body /
- producing blood cells /
- storing minerals /
- storing fat

(c) Any ***one set***: (2)

- Take in more calcium (1)
  for bone formation (1) /
- Take in more vitamin D (1)
  to help the absorption of calcium from food (1) /
- Do regular weight-bearing exercise (1)
  to stimulate the increase in bone density (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
	Analysing data	2
(b)	Understanding basic concepts	3
(c)	Applying concepts	2

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|!|BCELB0180000003|!| ★ **SI** **STSE**

(a) Give ***two*** reasons why bone is considered to be a living tissue. (2 marks)

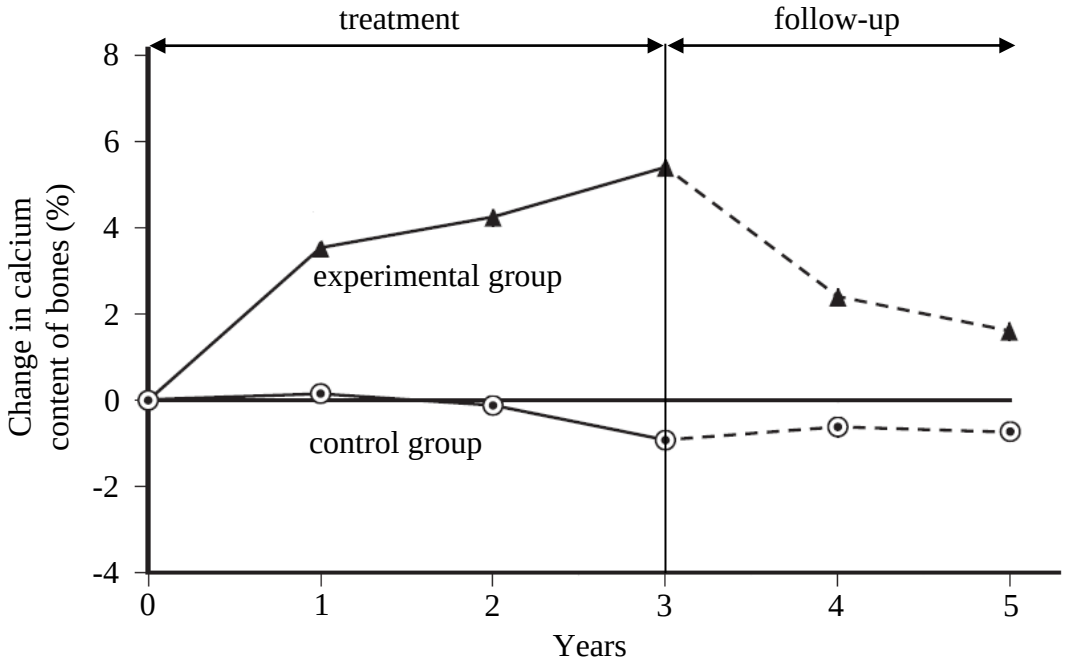
(b) State the properties of collagen and calcium phosphate in bone. (2 marks)

(c) Osteoporosis is a condition in which calcium is lost from the bones.

(i) Explain how osteoporosis leads to difficulty in moving around. (2 marks)

(ii) Osteoporosis is common in women after the menopause and is related to a decrease in oestrogen production. Doctors often prescribe oestrogen replacement therapy for these women.

The graph below shows the results of an investigation on the calcium content of the bones. Women in the experimental group were given pills containing oestrogen while women in the control group were given placebo pills. The treatment lasted for three years and scientists continued to monitor the change in calcium content the women's bones for two years further.



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Referring to the graph, describe the effect of oestrogen replacement therapy on preventing osteoporosis in post-menopausal women. (3 marks)

##

(a) Any ***two*** of the following: (2)

- Bone is capable of growth. /
- Bone is capable of repair. /
- Bone is capable of production of blood cells. /
- Cells in bone can respire. /

(Or other vital functions except irritability and movement)

(b) Collagen provides flexibility to bone. (1)

Calcium phosphate makes bone hard / rigid. (1)

(c) (i) Any ***two*** of the following: (2)

- The bone becomes weak / brittle. /
- The bone can break easily. /
- Pain arises when moving about.

(ii) Oestrogen increases calcium content of the bones. (1)

The effect is especially pronounced in the first year of treatment. / The increase is lesser in subsequent years. (1)

The effect decreases after the treatment is stopped. / The treatment needs to be continue to maintain the effect. (1)

(9 marks)

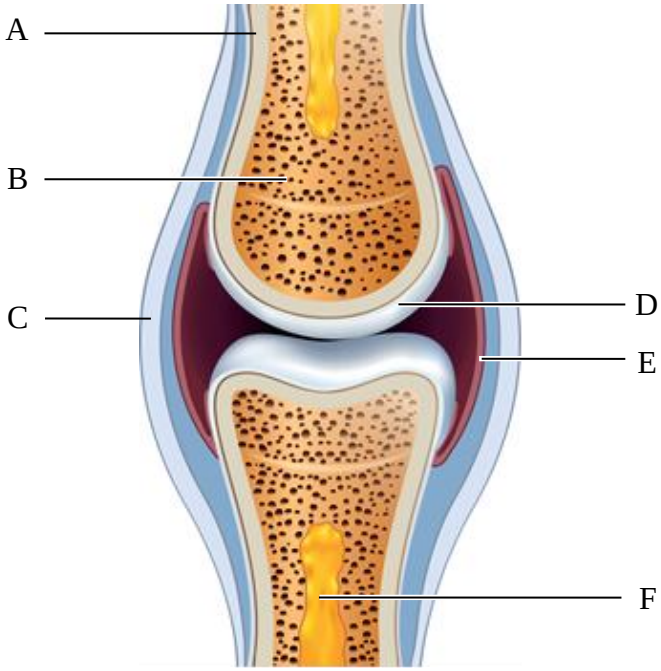
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)(i)	Applying concepts	2
(c)(ii)	Analysing data	3

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|!|BCELB0180000004|!| ★

The diagram below shows the structure of a joint.



(a) In which part of the human body can this type of joint be found? (1 mark)

(b) Using the letter from the diagram, indicate the site where red blood cells are produced. (1 mark)

(c) Give ***two*** differences between structures A and D. (2 marks)

(d) State and explain ***one*** feature of C which facilitates its function. (2 marks)

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(e) Name structure E and state its function. (2 marks)

_____

_____

(f) In a healthy joint, structure D is smooth. When a person has arthritis, structure D can become rough and uneven. Explain how arthritis will affect the functioning of the joint. (2 marks)

_____

_____

##

(a) Elbow / Knee / Finger (1)

(b) B (1)

(c) Any ***two*** of the following: (2)

- A is harder than D. / D is softer than A.
- A contains more minerals than D. / D contains fewer minerals than A.
- A is more rigid than D. / D is more elastic than A.
- A is supplied with blood while D is not.

(Or other reasonable answers)

(d) Any ***one set*** of the following: (2)

- C is tough. (1)
  It holds the bones together to prevent dislocation. (1) /
- C is elastic. (1)
  This allows the bones to move. (1)

(e) Synovial membrane (1)
It secretes synovial fluid, which reduces friction between the bones during movement / nourishes cartilage cells. (1)

(f) Movement at the joint will become difficult / painful (1)
due to increased friction / pieces of uneven cartilage rubbing against each other. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2
(e)	Understanding basic concepts	2
(f)	Applying concepts	2

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|!|BCELA0180000005|!| ★★ **Cross topic** **STSE**

The X-ray photograph below shows a dislocated shoulder joint.



(a) What type of joint does the shoulder joint belong to? Describe the range of movement this type of joint allows. (2 marks)

(b) Which structure of the joint was probably damage and caused the dislocation? (1 mark)

(c) A dislocated joint is usually swollen. Explain why. (4 marks)

(d) Suggest a way to reduce the swelling at the dislocated joint. (1 mark)

##

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(a) Ball-and-socket joint (1)

This type of joint allows movement of the bones in many planes / all directions. (1)

(b) Ligament (1)

(c) The damaged tissues release chemicals (histamine) (1)

which increases the permeability of capillary walls. (1)

This allows more blood plasma to move out from the blood capillaries to form tissue fluid. (1)

Excess tissue fluid accumulates at the joint (1)

and causes swelling.

(d) Applying an ice pack to the swollen area. (1)

(Or other reasonable answers)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Applying concepts	1
(c)	Applying concepts	4
(d)	Applying concepts	1

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|!|BCELB0180000006|!| ★

Figure 1 shows the pelvic girdle. Figure 2 shows the longitudinal section of the thigh bone.

Figure 1

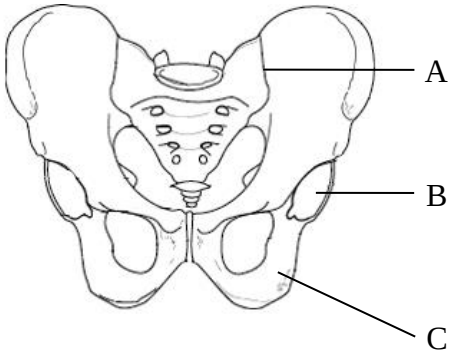
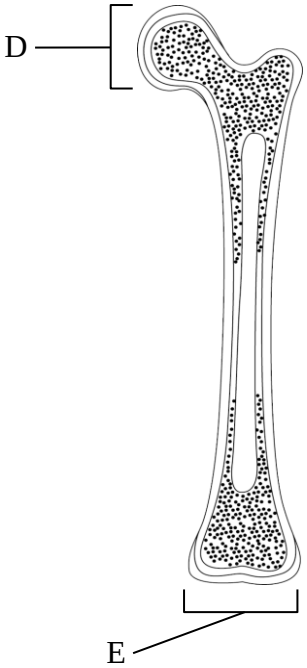


Figure 2



(a) (i) Between which two parts would a movable joint be formed? (1 mark)

(ii) Name the type of joint stated in (i) and describe the range of movement allowed by this type of joint. (2 marks)

(b) The cavities in the thigh bone contain bone marrow.

(i) Name the ***two*** types of bone marrow and give ***one*** function for each. (4 marks)

(ii) Suggest an advantage, apart from containing marrow, of a bone being hollow rather than solid. (1 mark)

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##

(a) (i) B and D (1)

(ii) Ball-and-socket joint (1)

This type of joint allows movement of the bones in many planes / all directions. (1)

(b) (i) Red bone marrow (1)

produces blood cells. (1)

Yellow bone marrow (1)

stores fat. (1)

(ii) This gives strength to the bone with the minimum of weight. (1)

(Or other reasonable answers)

(8 marks)

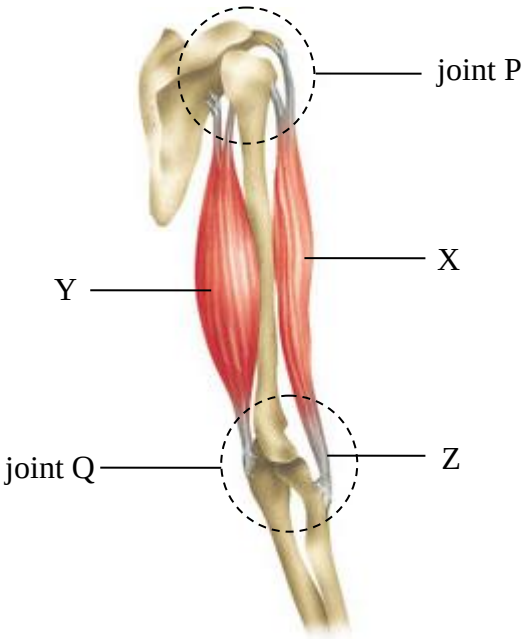
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
	Understanding basic concepts	1
(b)(i)	Understanding basic concepts	4
(b)(ii)	Critical thinking	1

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|!|BCELC0180000007|!|

The diagram below shows some bones and muscles in a human arm.



(a) (i) What types of movable joints are P and Q ? (2 marks)

(ii) State the difference in the range of movement allowed by joints P and Q. (2 marks)

(b) Muscles X and Y are a pair of antagonistic muscles.

(i) State which one is a flexor and which one is an extensor. (2 marks)

(ii) Describe the actions of muscles X and Y when the arm is bent at the elbow. (1 mark)

(c) Describe the role of structure Z when the arm is bent at the elbow. (2 marks)

##

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(a) (i) P: ball-and-socket joint (1)
Q: hinge joint (1)
(ii) Joint P allows movement of the bones in many planes / all directions. (1)
Joint Q allows movement of the bones in one plane. (1)
(b) (i) Muscle X is a flexor. (1)
Muscle Y is an extensor (1)
(ii) Muscle X contracts and muscle Y relaxes. (1)
(c) It transmits the pulling force generated by the contraction of muscle X (1)
to the bone (radius) of the lower arm. (1)

(9 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Understanding basic concepts	2
(b)(i)	Analysing data	2
(b)(ii)	Understanding basic concepts	1
(c)	Applying concepts	2

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|!|BCELA0180000008|!| ★★

Figure 1 shows some bones and muscles in a human arm. Figure 2 shows the transverse section of the arm along line PQ.

Figure 1

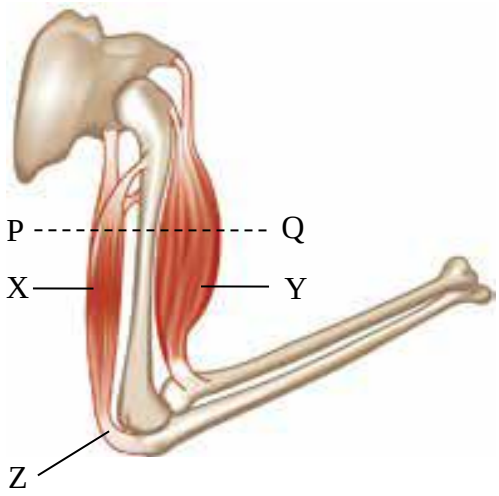
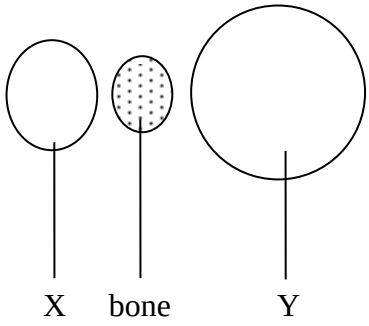


Figure 2



(a) Name muscles X and Y. (2 marks)

(b) Why are muscles X and Y described as an antagonistic muscle pair? (1 mark)

(c) With reference to Figure 2, redraw the muscles to show their shapes if the arm is straightened. (2 marks)

(d) Name structure Z. (1 mark)

(e) Explain the elasticity of structure Z in relation to its functions. (2 marks)

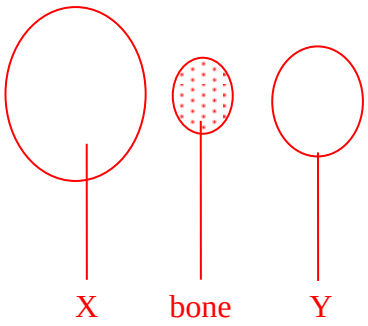
##

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(a) X: triceps (1)
Y: biceps (1)

(b) When producing movements, one muscle contracts, the other relaxes. / The contraction of Y bends the arm and the contraction of X extends the arm. (1)

(c) Larger muscle X (1)
Smaller muscle Y (1)



(d) Tendon (1)

(e) It is inelastic. (1)
This allows effective transmission of the pulling force generated by muscle contraction to the bones. (1)

(8 marks)

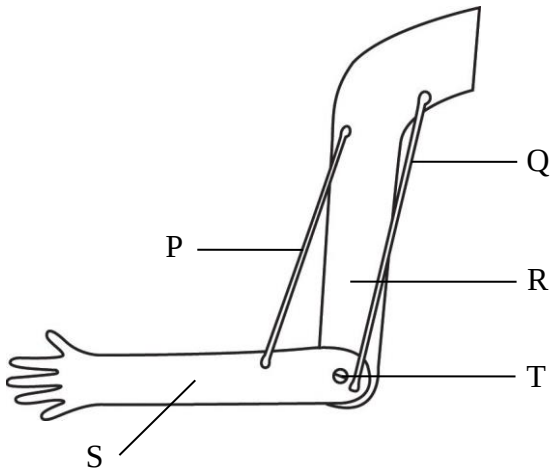
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	1
(c)	Presenting information	2
(d)	Analysing data	1
(e)	Understanding basic concepts	2

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|!|BCELB0180000009|!| ★

The diagram below shows a model used to demonstrate movement of the forearm. P and Q are rubber bands. R and S are acrylic boards. T is a screw.



(a) P and Q represent muscles in the arm.

(i) State which rubber band will shorten when S is lifted. (1 mark)

(ii) State which rubber band will shorten when S is lowered. (1 mark)

(iii) What is the term used for describing this pair of muscles? (1 mark)

(b) Name the type of movable joint represented by T. (1 mark)

(c) State ***two*** ways in which the model cannot truly represent the human arm. (2 marks)

##

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(a) (i) P (1)

(ii) Q (1)

(iii) Antagonistic / Opposing muscles (1)

(b) Hinge joint (1)

(c) Any ***two sets*** of the following: (2, 2)

In the model	In actual case
The arm is bent when at its resting position. (1)	The arm is straightened when at its resting position. (1)
Movement of the forearm (board S) is caused by external force. (1)	Movement of the forearm is caused by the contraction of the biceps. (1)
Rubber bands are directed attached to the acrylic boards. (1)	Muscles are attached to bones by inelastic tendons. (1)

(8 marks)

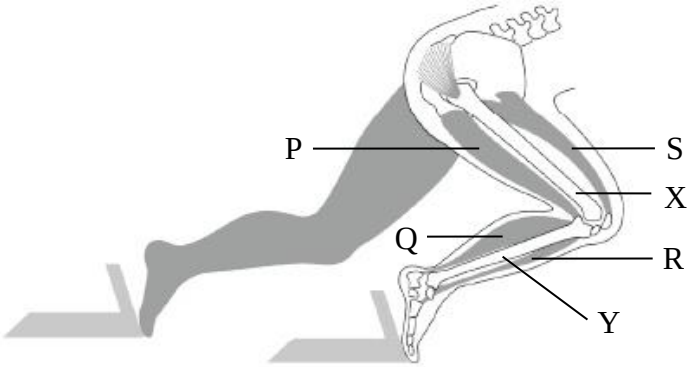
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(a)(iii)	Understanding basic concepts	1
(b)	Analysing data	1
(c)	Critical thinking	4

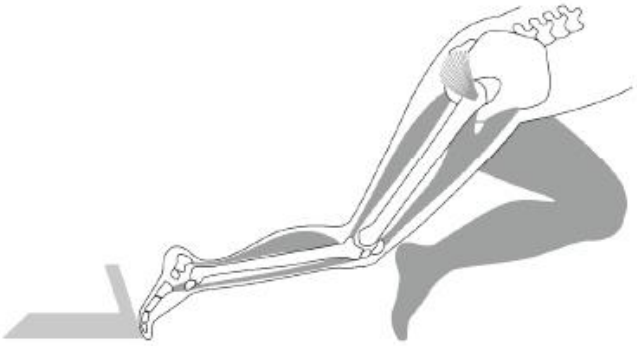
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|!|BCELA0180000010|!| ★★

The diagram below shows the bones and muscles in the right leg of an athlete as he leaves the starting block at the start of a race.



position 1



position 2

(a) How does an antagonistic muscle pair work? (2 marks)

_____

_____

(b) Using the letters in the diagram, indicate ***two sets*** of antagonistic muscles. (2 marks)

_____

_____

(c) Which muscle(s) will contract as the athlete moves from position 1 to position 2? (1 mark)

_____

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(d) Name the tissues by which bones X and Y are held together. (1 mark)

(e) At a joint, the ends of the bones are covered by cartilage.

(i) Describe ***one*** other way in which friction at the joint is reduced. (1 mark)

(ii) Problems with knee joint, such as cartilage damage, are common in athletes. Explain what the damage is and how it would affect the athlete. (3 marks)

##

(a) An antagonistic muscle pair works together to bring about movement. (1)
When one muscle of the pair contracts, the other relaxes. (1)

(b) P and S (1)
Q and R (1)

(c) Q and S (1)

(d) Ligaments (1)

(e) (i) Synovial fluid which serves as a lubricant is present in the joint. (1)

(ii) Cartilage would wear away / become thinner. (1)
Movement at the joint will become difficult / painful (1)
due to increased friction / pieces of uneven cartilage rubbing against each other. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	2
(c)	Analysing data	1
(d)	Understanding basic concepts	1
(e)(i)	Understanding basic concepts	1
(e)(ii)	Applying concepts	3

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|!|BCELA0180000011|!| ★★

Figure 1 shows a man sitting and using a laptop. Figure 2 shows the muscles and bones in his right leg.



Figure 1

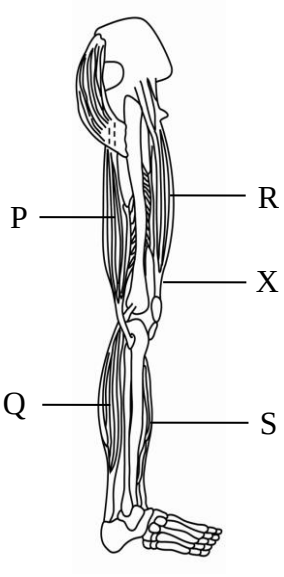


Figure 2

(a) Which muscle(s) of the right leg (P, Q, R or S) is/are contracting when the man is sitting in the posture as shown in Figure 1? (1 mark)

(b) With reference to their relative actions, what is the term used to describe muscles P and R? Explain your answer. (2 marks)

(c) (i) Name structure X. (1 mark)

(ii) With reference to the physical properties of X, explain how it is adopted to the role it plays as the man stands up. (3 marks)

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(d) Explain ***two*** harmful effects of a bad sitting posture such as the one shown in Figure 1. (4 marks)

##

(a) P and S (1)

(b) Antagonistic / Opposing muscles (1)

Contraction of muscle P brings about bending of the leg while that of muscle R brings about straightening of the leg. (1)

(c) (i) Tendon (1)

(ii) It is tough and inelastic, (1)

so that it can transmit the force (1)

generated from the contraction of muscle R (1)

to pull onto the bones of the lower leg.

(d) Any ***two sets*** of the following: (2, 2)

- It may put tension in muscles (1)
  causing muscle pain in the neck / shoulder / back. (1)
- It may put excess pressure on intervertebral disc / the vertebral column (1)
  causing chronic back pain / deformation of the vertebral column. (1)
- Abdominal organs are being pressed on (1)
  resulting in poor digestion. (1)

(Or other reasonable answers)

(11 marks)

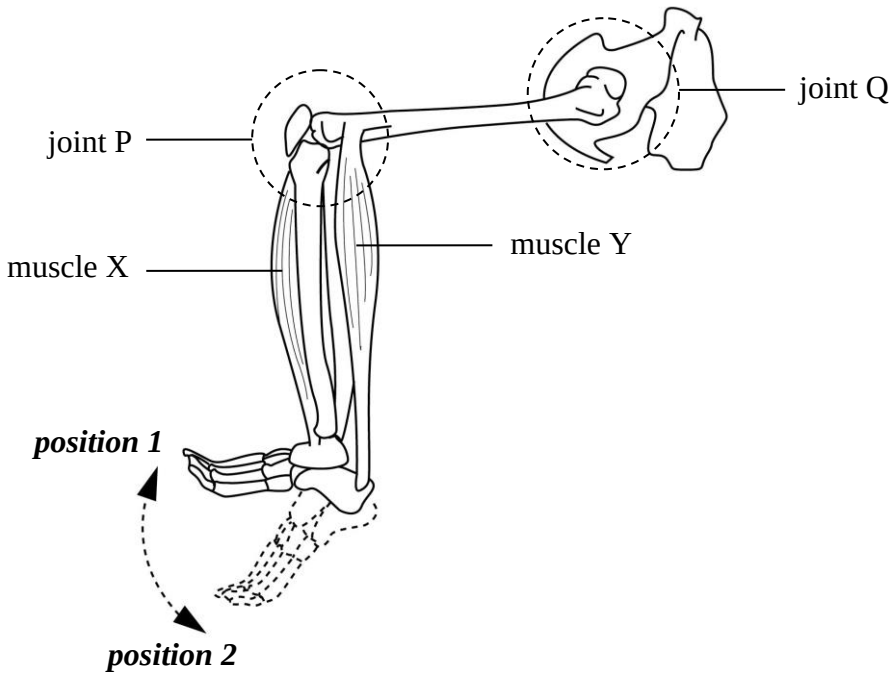
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
	Applying concepts	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1
	Applying concepts	2
(d)	Applying concepts	4

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|!|BCELB0180000012|!| ★

The diagram below shows the bones and some muscles of the leg of a man who is sitting.



(a) (i) What types of movable joints are joints P and Q? (2 marks)

_____

_____

(ii) Give ***one*** other example of each type of joint in (i). (2 marks)

_____

_____

(b) Name ***one*** elastic structure in the joint. State ***one*** function of this structure in facilitating movement at the joint. (2 marks)

_____

_____

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(c) Muscles X and Y work in a pair.

(i) What is the term used for describing this pair of muscles? (1 mark)

_____

(ii) Describe how the actions of these two muscles bring about the movement of the foot from position 1 to position 2. (2 marks)

_____

_____

##

(a) (i) Joint P: hinge joint (1)

Joint Q: ball-and-socket joint (1)

(ii) Example of joint P: elbow joint (1)

Example of joint Q: shoulder joint (1)

(b) Any ***one set*** of the following: (2)

- Ligament (1)
  It holds the bones in position / prevents dislocation. (1) /
- Cartilage (1)
  It absorbs shock / reduces friction between the bones during movement / prevents the bones from wearing. (1)

(c) (i) Antagonistic / Opposing muscles (1)

(ii) Muscle Y contracts (1)

and muscle X relaxes. (1)

(9 marks)

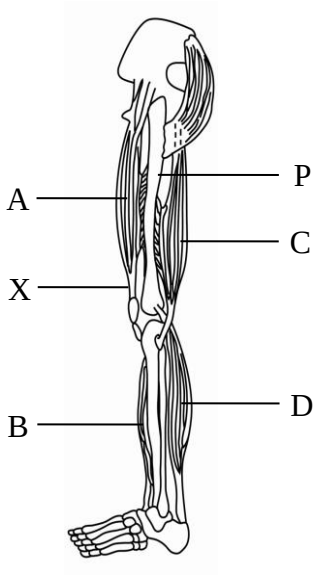
##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	2

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|!|BCELC0180000013|!|

The diagram below shows the bones and muscles in a human leg.



(a) Using the letters provided, indicate the extensor and flexor involved in straightening and bending the leg at the knee. Explain your answer. (4 marks)

(b) A student said 'Antagonistic muscles work in pairs, where one muscle contracts and the other expands'.
Comment on the statement. (2 marks)

(c) What feature of structure X ensures that all the force from a muscle contraction is transmitted to the bone? (1 mark)

(d) (i) Bone P is easily broken in people suffering from osteoporosis. What kind of mineral is lost from the bones in people suffering from osteoporosis? (1 mark)

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(ii) Apart from increasing dietary intake of the mineral, suggest, with reason, another way to reduce the chance of getting osteoporosis. (2 marks)

_____

_____

##

(a) Extensor: A (1)
Contraction of A straightens the leg at the knee. (1)
Flexor: C (1)
Contraction of C bends the leg at the knee. (1)

(b) Muscles can only work by contracting and never by expanding. (1)
When one muscle in an antagonistic pair contracts, the other in the pair relaxes. (1)

(c) It is inelastic. (1)

(d) (i) Calcium (1)

(ii) Any ***one set*** of the following: (2)

- Take in more vitamin D (1)
  to help the absorption of calcium from food (1) /
- Do regular weight-bearing exercise (1)
  to stimulate the increase in bone density (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Understanding basic concepts	2
(b)	Critical thinking	1
	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)(i)	Understanding basic concepts	1
(d)(ii)	Applying concepts	2

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|!|BCELA0180000014|!| ★★ **STSE**

Osteoarthritis is a joint disease in which the cartilage covering the ends of the bones breaks down.

(a) The most common symptom of osteoarthritis is joint pain. Suggest how osteoarthritis causes joint pain. (1 mark)

(b) The table below shows the numbers of males and females, from different age groups in the UK, diagnosed with osteoarthritis of the hip or knee.

Age group (years)	**Gender**	**Number of individuals diagnosed with osteoarthritis**	
		Hip	**Knee**
25–44	male	9000	5400
	female	9000	21,600
45–64	male	15,500	18,200
	female	15,500	72,800
65 or above	male	77,000	74,000
	female	77,000	296,000

(i) Based on the data in the table, describe the relationship between the occurrence of arthritis in the hips and knees with age and gender. (3 marks)

(ii) Over the last decade, there has been a significant increase in the number of people with osteoarthritis. Suggest ***one*** reason for this increase. (1 mark)

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(c) In severe cases of osteoarthritis, joint replacement may be needed. Two artificial joints, X and Y, are shown below.



X



Y

(i) Which artificial joint, X or Y, can be used to replace the hip joint? (1 mark)

_____

(ii) State ***two*** criteria for choosing the materials to make artificial joints. (2 marks)

_____

_____

_____

##

(a) Pain arises due to increased friction / as pieces of uneven cartilage rubbing against each other. (1)

(b) (i) The occurrence of both types of osteoarthritis increase with age in both males and females. (1)

In all age groups, osteoarthritis affects males and females the same in the hips. (1)

In all age groups, osteoarthritis affects females more in the knees. (1)

(ii) There has been an increase in the occurrence obesity / an increase in the aging population.

(Or other reasonable answers) (1)

(c) (i) Y (1)

(ii) Any ***two*** of the following: (2)

- Non-toxic /
- High strength to resist bending and compression /
- Light in weight /
- Resistant to corrosion

(Or other reasonable answers)

_____

(8 marks)

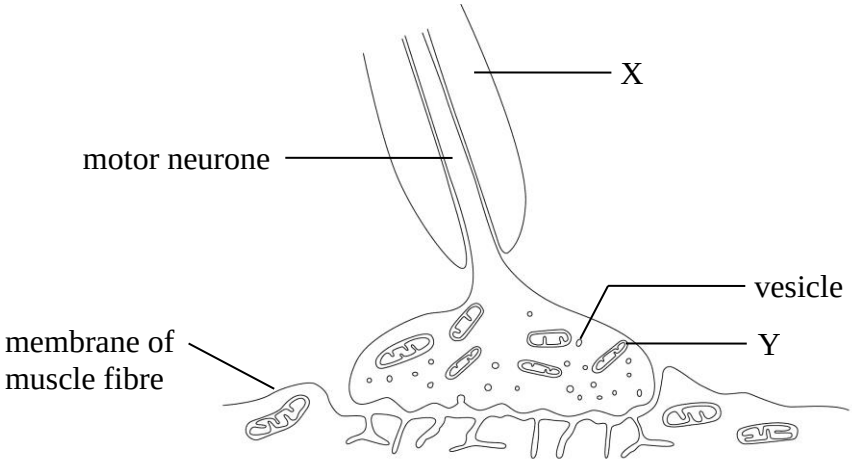
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##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)(i)	Analysing data	3
(b)(ii)	Critical thinking	1
(c)(i)	Analysing data	1
(c)(ii)	Critical thinking	2

|!|BCELB0180000015|!| ★

The diagram below shows a neuromuscular junction.



(a) Name X and state its function. (2 marks)

(b) What is present in the vesicle? (1 mark)

(c) Describe the events that happen in the neuromuscular junction when a nerve impulse arrives at the ending of the motor neurone. (4 marks)

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(d) Explain the significance of organelle Y being abundant in the ending of the motor neurone and in the muscle fibre. (2 marks)

##

(a) Myelin sheath (1)
It protects the nerve fibre. / It insulates the nerve fibre. / It speeds up the transmission of nerve impulses. (1)

(b) Neurotransmitter (1)

(c) The arrival of a nerve impulse at the ending of the motor neurone stimulates the release of neurotransmitters. (1)
The neurotransmitters diffuse across the cleft in the neuromuscular junction (1)
and bind to receptor sites on the membrane of the muscle fibre (1)
to initiate muscle contraction. (1)

(d) Organelle Y is abundant in the motor neurone to provide energy for the formation of neurotransmitters / formation of vesicles / movement of vesicles / release of neurotransmitters. (1)
Organelle Y is abundant in the muscle fibre to provide energy for contraction. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	4
(d)	Applying concepts	2

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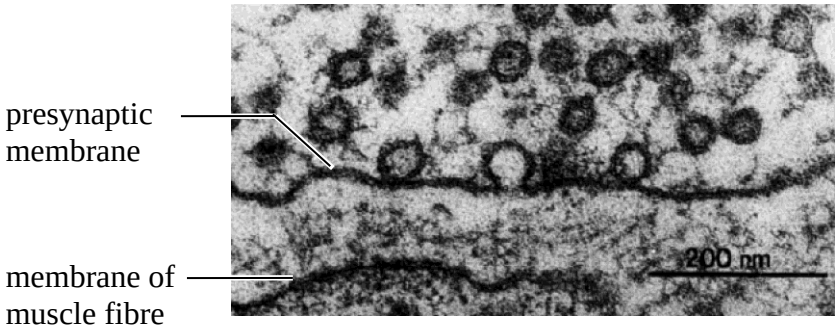
|!|BCELA0180000016|!| ★★ **STSE**

Photomicrograph 1 shows a neuromuscular junction. Photomicrograph 2 shows the cleft in the neuromuscular in a higher magnification.

Photomicrograph 1



Photomicrograph 2



(a) What type of neurone is X? Explain your answer. (2 marks)

(b) Referring to photomicrograph 1, describe ***one*** observable feature of neurone X and explain how it is related to the functioning of the neuromuscular junction. (2 marks)

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(c) Botulism is form of food poisoning caused by the ingestion of a toxin produced by the bacterium *Clostridium botulinum*. The toxin acts by blocking the functioning of the presynaptic membrane, resulting in paralysis. Based on this information and photomicrograph 2, explain how this toxin causes paralysis.

(4 marks)

##

(a) Motor neurone (1)

It is connected to an effector (muscle fibre in this case). (1)

(b) There are numerous vesicles in the ending of neurone X. (1)

The vesicles contain neurotransmitters which diffuse across the cleft neuromuscular junctiont. (1)

(c) The toxin blocks the release of neurotransmitters into the cleft. (1)

There will be no neurotransmitters to stimulate the membrane of the muscle fibre. (1)

The membrane of the muscle fibre will not generate nerve impulses (1)

and the muscle will not contract. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Understanding basic concepts	1
	Understanding basic concepts	1
(d)	Applying concepts	4

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